

Claude Code + n8n Complete Tutorial

From Zero to Automated Workflow Creation

What We Are Building

You will have a system where:

- You open VS Code
- Type **claude** in the terminal
- Describe what you want automated
- Claude Code **automatically builds, validates, deploys and tests** the workflow in your n8n

No manual JSON editing. No copy-pasting. Just describe and it is done.

How Everything Connects (The Big Picture)

Layer	Component	Role
 You	Describe what you want	Input
 Claude Code	Runs in VS Code terminal	The brain
 CLAUDE.md	Your rules + n8n URL + API key	Instructions
 Skills	Expert n8n knowledge	Knowledge
 MCP Tools	Connects to your n8n	The hands
 n8n on Hetzner	https://n8n1.mindvault.live	Your server
 Result	Workflow created, tested, live!	Output

Why this architecture works:

- CLAUDE.md tells Claude YOUR specific setup and rules
- Skills gives Claude expert n8n knowledge built-in
- MCP gives Claude actual hands to touch your n8n
- Together, Claude builds workflows like an expert who already knows your system

Prerequisites

- Windows PC with VS Code installed
 - Claude Code installed globally — run: **npm install -g @anthropic-ai/claude-code**
 - n8n running on Hetzner via Coolify at <https://n8n1.mindvault.live>
 - Anthropic API key from console.anthropic.com
 - Node.js installed — check by running: **node --version**
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STEP 1 — Get Your n8n API Key

Why: Claude Code needs this key to authenticate with your n8n and create workflows on your behalf. Without it, Claude can see your n8n but cannot touch it.

1. Go to <https://n8n1.mindvault.live>
2. Click **Settings** at the bottom left
3. Click **API**
4. Click **Create API Key**
5. Give it a name like "claude-code"
6. Copy the key — it looks like eyJhbGci...

⚠ Important: Never share this key publicly. Store it only in your CLAUDE.md and terminal commands.

STEP 2 — Create Your Project Folder

Why: Claude Code reads CLAUDE.md automatically from whatever folder you open in VS Code. Having one dedicated folder means your config loads every session without any extra work.

Your working folder path is:

C:\Users\user\OneDrive\Desktop\DATA\AGENTIC WORKFLOW\n8n

1. Open VS Code
2. Click **File → Open Folder** and select the **n8n** folder

Always open the **n8n** folder specifically — that is where CLAUDE.md lives.

This folder contains:

- **CLAUDE.md** — your permanent config file, never delete this
 - **Workflow JSON files** — these get created automatically over time
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STEP 3 — Create CLAUDE.md

Why: This is the most important file. Claude Code reads it at the start of every session. It tells Claude your n8n URL, API key, rules to follow, and the exact process to use when creating workflows. Without this, Claude has to guess your setup every time.

Create a new file called **CLAUDE.md** in your n8n folder and paste the content below into it. Replace YOUR-REAL-API-KEY-HERE with your actual key.

CLAUDE.md content — copy from here:

My n8n Setup

Instance:

- URL: <https://n8n1.mindvault.live>
- API Key: YOUR-REAL-API-KEY-HERE

API Usage:

- Endpoint: POST /api/v1/workflows
- Auth: Authorization Bearer YOUR-REAL-API-KEY-HERE
- Always activate workflow after importing

Workflow Rules:

- Use webhook trigger unless I specify otherwise
- Add sticky notes explaining each section
- Use descriptive node names, never leave default names like "HTTP Request 1"
- Name workflows clearly based on what they do

Workflow Creation Process — ALWAYS follow these steps:

Step 1 — Check environment:

- Run `n8n_health_check` to check n8n version

- Run `search_nodes` to verify required nodes exist in my instance
- Run `n8n_manage_credentials` to check available credentials
- Configure all nodes for the exact version found

Step 2 — Build the workflow:

- Use only nodes confirmed to exist in my n8n
- Reference credentials by their exact names from n8n
- Name every node descriptively
- Add sticky notes explaining each major section

Step 3 — Validate before deploying:

- Run `n8n_validate_workflow` on the workflow JSON
- Fix any validation errors before proceeding

Step 4 — Deploy:

- Run `n8n_create_workflow` to deploy to my n8n
- Activate the workflow automatically after creating

Step 5 — Test:

- Run `n8n_test_workflow` to test it
- Check output of each node carefully
- If any node fails, read the full error message

Step 6 — Fix and retry:

- If something fails, run `n8n_autofix_workflow`
- Fix the specific broken node
- Re-validate, re-deploy, re-test
- Keep retrying until workflow runs successfully
- If you need credentials info or any input from me, ask me

Step 7 — Confirm:

- Tell me the workflow is live and active
- Give me the webhook URL if applicable
- Explain what each node does in simple terms

My Stack:

- n8n is self-hosted on Hetzner via Coolify running Docker
- Prefer Gemini or OpenAI for AI nodes
- Use Google Drive for document outputs
- Use Gmail for email nodes

End of CLAUDE.md content

Save the file after pasting.

STEP 4 — Open Terminal in VS Code

Why: Claude Code runs as a CLI tool in your terminal. The VS Code integrated terminal is the most convenient place because you can see your files and terminal side by side.

1. In VS Code press **Ctrl + backtick** (the backtick key is top left of your keyboard, below Escape)
 2. A terminal opens at the bottom of VS Code
 3. Confirm you are in the right folder by typing **pwd** and pressing Enter
 4. It should show your n8n folder path
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STEP 5 — Add the n8n MCP Server

Why: MCP (Model Context Protocol) gives Claude Code actual tools to interact with your n8n. Without MCP, Claude can only write text. With MCP, Claude has tools like `n8n_create_workflow`, `n8n_test_workflow`, and `n8n_autofix_workflow` — real actions it can perform. Think of MCP as giving Claude hands instead of just a brain.

This is a one-time setup. You never need to run this again unless you change your API key.

In the PowerShell terminal — not inside Claude Code, just the regular terminal — type the following command as **one single long line with no line breaks**:

```
claude mcp add n8n-mcp -e MCP_MODE=stdio -e LOG_LEVEL=error -e  
DISABLE_CONSOLE_OUTPUT=true -e N8N_API_URL=https://n8n1.mindvault.live -e  
N8N_API_KEY=YOUR-REAL-API-KEY-HERE -- npx n8n-mcp
```

Replace YOUR-REAL-API-KEY-HERE with your actual key. Paste the whole thing on one line then press Enter.

You should see the message: **MCP server n8n-mcp added**

STEP 6 — Start Claude Code

Why: Now that MCP is configured and CLAUDE.md is ready, starting Claude Code will automatically load everything — your config, MCP tools, and all skills.

In the terminal, type **claude** and press Enter.

You will see Claude Code start up showing:

- Your folder name
 - Model: Sonnet 4.6
 - CLAUDE.md shown at the bottom of the chat panel
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STEP 7 — Install n8n Skills Plugin

Why: Skills are compressed expert knowledge. The n8n skills plugin gives Claude deep knowledge about n8n node syntax, expressions, workflow patterns, and common mistakes — all in about 570 tokens. Without skills, Claude would guess how n8n nodes work and make mistakes. With skills, it builds workflows correctly the first time.

Inside the Claude Code chat panel, type:

```
| /plugin marketplace add czlonkowski/n8n-skills
```

Then type:

```
| /plugin install
```

You should see 7 skills load:

- **n8n-workflow-patterns** — knows the 5 core workflow patterns
- **n8n-mcp-tools-expert** — knows how to use MCP tools correctly
- **n8n-validation-expert** — knows how to validate and fix workflows
- **n8n-node-configuration** — knows correct node settings
- **n8n-expression-syntax** — knows n8n expression language
- **n8n-code-javascript** — knows JavaScript for code nodes
- **n8n-code-python** — knows Python for code nodes

These load automatically every session. You never need to install them again.

STEP 8 — Verify Everything is Connected

Why: Before building real workflows, confirm Claude Code can actually reach your n8n. This saves you from debugging later.

Inside Claude Code, type:

┃ Run `n8n_health_check` and tell me my n8n version and status

Claude should respond with your n8n version number and confirm it is connected.

Then check the full context by typing: **/context**

You should see MCP tools loaded, Skills loaded, and CLAUDE.md in the context section.

STEP 9 — Create Your First Workflow

Why: Now everything is ready. Just describe what you want and Claude handles the rest — checking nodes, building, validating, deploying, testing, and fixing errors automatically.

Try typing this inside Claude Code:

┃ Create a simple n8n workflow that has a webhook trigger, receives a name field in the body, and returns a JSON response saying Hello followed by the name. Follow all steps in CLAUDE.md.

Claude Code will automatically:

1. Check your n8n version using `n8n_health_check`
 2. Verify the webhook node exists using `search_nodes`
 3. Build the workflow JSON using skills knowledge
 4. Validate it using `n8n_validate_workflow`
 5. Deploy it using `n8n_create_workflow`
 6. Test it using `n8n_test_workflow`
 7. Fix anything broken using `n8n_autofix_workflow`
 8. Tell you the webhook URL when done
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STEP 10 — Add MCP to VS Code Extension Chatbox (Optional)

Why: The VS Code extension chatbox is better for uploading files such as PDFs, images, and documents because it has a + button for attachments. To use it for workflow creation you need to add MCP there too.

1. Press **Ctrl + Shift + P**
2. Type **Claude: Open User MCP Settings** and click it
3. A JSON settings file opens
4. Replace the entire contents with the following — put it all as one block, replacing YOUR-REAL-API-KEY-HERE with your key:

```
{ "mcpServers": { "n8n-mcp": { "command": "npx", "args": ["n8n-mcp"], "env": {  
  "MCP_MODE": "stdio", "LOG_LEVEL": "error", "DISABLE_CONSOLE_OUTPUT": "true",  
  "N8N_API_URL": "https://n8n1.mindvault.live", "N8N_API_KEY": "YOUR-REAL-API-KEY-  
  HERE" } } } }
```

5. Save the file and restart VS Code
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Your Daily Routine

Every time you want to create a workflow:

1. Open VS Code
 2. Click **File → Open Folder** and select the **n8n** folder at:
C:\Users\user\OneDrive\Desktop\DATA\AGENTIC WORKFLOW\n8n
 3. Press **Ctrl + backtick** to open the terminal
 4. Type **claude** and press Enter
 5. Describe the workflow you want
 6. Done — Claude builds and deploys it automatically
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How to Upload Documents and Images

In the VS Code extension chatbox:

1. Click the + button at the bottom left of the chatbox
2. Select your PDF, image, or document

- 3. Type your prompt alongside it, for example: Build an n8n workflow based on this document
- 4. Claude analyzes the file and creates the workflow

In the terminal Claude Code: Reference the file path directly in your message, for example:

```
Analyze this document and build an n8n workflow based on it: C:\Users\user\Desktop\my-process.pdf
```

Your Folder Structure Over Time

Path	What it is
n8n\	Always open THIS in VS Code
n8n\CLAUDE.md	Your config — never delete
n8n\workflow-name.json	Auto-created by Claude

Old workflow files do not cost tokens. Claude Code only reads files you specifically ask it to read.

Understanding Token Usage

Category	Tokens	Percentage
System prompt	6,800	3.4%
System tools	8,000	4.0%
MCP tools	995	0.5%
Skills	570	0.5%
Your messages	3,000	1.5%
Free space	147,000	73.5%

MCP and Skills together use only 1% of your 200,000 token limit. You have 73.5% free for actual work. Skills and MCP save tokens because you do not need to type long explanations every session.

Troubleshooting

"MCP server n8n-mcp already exists" This is fine — it was already added. Just type **claude** and proceed normally.

Claude cannot connect to n8n

- Check your API key in CLAUDE.md is the correct current key
- Go to n8n Settings → API and verify the key is still active
- Check Coolify dashboard — confirm n8n is running and showing green status

Workflow was created but not showing in n8n UI

- Refresh your n8n browser tab
- Ask Claude: "Check if the workflow was actually created in n8n"

Skills not loading

- Type **/plugin install** again inside Claude Code

PowerShell gives an error about -- or backslash

- Always run the MCP command as one single line with no line breaks

Quick Reference Card

Task	What to type
Start Claude Code	claude
Check what is loaded	/context
Install skills	/plugin marketplace add czlonkowski/n8n-skills
Check MCP servers	claude mcp list
Exit Claude Code	Ctrl + C
Resume last session	claude --resume

Example Prompts to Try

Email monitoring: Create an n8n workflow that monitors a Gmail inbox for emails with "invoice" in the subject and saves them to Google Sheets. Follow all steps in CLAUDE.md.

Lead qualification: Build a webhook workflow that receives customer data, qualifies the lead using AI, and sends a Slack notification with the result. Follow all steps in CLAUDE.md.

Weekly report: Create a scheduled workflow that runs every Monday at 9am, pulls data from Google Sheets, generates an AI summary, and emails it to me. Follow all steps in CLAUDE.md.

Document-based workflow: Read this PDF and create an n8n workflow that automates the process described in it. Follow all steps in CLAUDE.md. Then attach your PDF using the + button.

Setup is one-time only. After completing Steps 1 through 7 once, every new workflow is just: open VS Code, open the n8n folder, type claude, describe what you want.